

3	(a)	A real gas approaches ideal gas behaviour at <u>high temperature</u> and <u>low pressure</u>. [B1] At high temperature, the <u>intermolecular forces are negligible</u> as the gas molecules moving very rapidly. [B1] At low pressure, <u>total volume of the gas molecules is negligible (or particles are far apart)</u> compared to the <u>volume of the container</u>. [B1]
	(b)(i)	$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$ $\frac{(4.45 \times 10^3)(0.28)}{350} = \frac{(2.42 \times 10^4)(0.12)}{T} \quad [\text{C1}]$ $T = 816 \text{ K} \quad [\text{A1}]$ Or use $pV = nRT$
	(b)(ii)	$\Delta U = \frac{3}{2} n R \Delta T$ $= \frac{3}{2} (0.428)(8.31)(350 - 816) \quad [\text{M1}]$ $= -2490 \text{ J}$ $\Delta U = Q + W$ $\therefore W = 0 \quad [\text{M1}]$ $\Delta U = Q \quad [\text{M1}]$ $= -2490 \text{ J} \quad [\text{A0}]$ Hence, heat removed = 2490 J [A0]
	(b)(iii)	For adiabatic process, Q is zero. For a cyclic process, $\Delta U_{A \rightarrow B \rightarrow C} = 0$ $\Delta U_A + 0 + (-2490) = 0 \quad [\text{M1}]$ $\Delta U_A = Q + W_A$ $W_A = 2490 \text{ J} \quad [\text{A1}]$

4	(a)	The total electric potential is negative for all points between them	M1
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